

Riemann surfaces problem set 4

1. Let f be a meromorphic function on a Riemann surface X such that f is not identically zero. Show that the 1-form $\omega = df/f$ is a meromorphic 1-form on X whose only poles are simple poles at the zeroes and poles of f , and

$$\text{Res}(df/f, p) = \text{ord}_p(f)$$

for all $p \in X$.

2. Let X be a compact Riemann surface of genus zero and let ω be a meromorphic 1-form on X . Show that $\omega = df$ for some meromorphic function f on X if and only if all residues of ω are equal to zero.

3. Let X be a Riemann surface, let $p \in X$, and let z be a local coordinate of X in a neighbourhood of p . Define an endomorphism $J_p : T_p X \rightarrow T_p X$ by $J_p := (dz_p)^{-1} \circ M_i \circ dz_p$ where $M_i : \mathbb{C} \rightarrow \mathbb{C}$ denotes multiplication by i .

a) Show that the endomorphism J_p is independent of the choice of local coordinate z near p .

b) Show that a C^1 function $f : U \subset X \rightarrow \mathbb{C}$ defined on an open subset $U \subset X$ is holomorphic if and only if $df_p \circ J_p = i \cdot df_p$ for all $p \in U$.

c) Given a local coordinate $z = x + iy$ near $p \in X$, let $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}$ denote the corresponding basis of $T_p X$. Show that $J_p(\frac{\partial}{\partial x}) = \frac{\partial}{\partial y}$, $J_p(\frac{\partial}{\partial y}) = -\frac{\partial}{\partial x}$.

d) Let $J_p^* : T_p^* X \rightarrow T_p^* X$ be defined by $J_p^* \alpha = \alpha \circ J_p$ for $\alpha \in T_p^* X$. Show that for any local coordinate $z = x + iy$ near p we have $J_p^* dx = -dy$, $J_p^* dy = dx$.

4. Let X, Y be Riemann surfaces and let $J_X : TX \rightarrow TX, J_Y : TY \rightarrow TY$ denote the canonical bundle endomorphisms defining the complex structures on the tangent spaces of X and Y respectively. Show that a C^1 map $f : X \rightarrow Y$ is holomorphic if and only if $df \circ J_X = J_Y \circ df$.

5. Let X be a Riemann surface, and let ω be a C^∞ complex 1-form on X . Show that the following are equivalent:
- a) ω is a holomorphic 1-form.
 - b) For any local coordinate $z : U \subset X \rightarrow \mathbb{C}$, $\omega = g dz$ on U where g is a holomorphic function on U .
 - c) ω is a closed $(1, 0)$ -form.
6. Recall that a function $f : X \rightarrow \mathbb{C}$ on a Riemann surface X is said to be harmonic if $f \circ z^{-1}$ is harmonic for any local coordinate z on X .
- a) Show that if a real-valued harmonic function $f : X \rightarrow \mathbb{R}$ on X attains its maximum then it is constant.
 - b) Show that a smooth function $f : X \rightarrow \mathbb{C}$ is harmonic if and only if $d * df = 0$, where $*$ denotes the Hodge star operator acting on 1-forms.
 - c) Show that if X is simply connected, then any real-valued harmonic function u on X has a harmonic conjugate v , i.e. there exists a real-valued harmonic function v on X such that the function $f = u + iv$ is holomorphic.
7. Recall that a 1-form on a Riemann surface X is said to be harmonic if locally it is given by the derivative of a harmonic function.
- a) Show that a smooth 1-form ω on X is harmonic if and only if $d\omega = d * \omega = 0$.
 - b) Show that any harmonic 1-form on X can be written uniquely as the sum of a holomorphic 1-form and an anti-holomorphic 1-form.
 - c) Let ω be a harmonic 1-form on a compact Riemann surface X . Show that if ω is exact, then $\omega \equiv 0$.
8. Show that a smooth 1-form ω on a Riemann surface X is harmonic if and only if
- $$\int_X \omega \wedge df = \int_X \omega \wedge *df = 0$$
- for all $f \in C_c^\infty(X)$.

9. Let ω be a smooth 1-form on a Riemann surface X . Show that ω is exact if and only if $\int_X \omega \wedge \alpha = 0$ for all smooth closed 1-forms α with compact support.
10. Let X be a compact Riemann surface of genus g and let $a_1, \dots, a_g, b_1, \dots, b_g$ be simple closed curves on X forming a canonical homology basis of X (so the algebraic intersection numbers satisfy $a_i \cdot b_j = \delta_{ij}$).

a) Show that for any two closed 1-forms α, β on X we have

$$\int_X \alpha \wedge \beta = \sum_{j=1}^g \left(\int_{a_j} \alpha \int_{b_j} \beta - \int_{b_j} \alpha \int_{a_j} \beta \right)$$

b) Show that if a holomorphic 1-form ω on X satisfies $\int_{a_j} \omega = 0$ for all $j = 1, \dots, g$, then $\omega \equiv 0$.

c) Show that there exist holomorphic 1-forms $\omega_1, \dots, \omega_g$ on X such that $\int_{a_j} \omega_i = \delta_{ij}$.

11. Let X be a compact Riemann surface of genus one, let a, b be simple closed curves on X forming a canonical homology basis of X (so the algebraic intersection number $a \cdot b = 1$). Let ω be a holomorphic 1-form on X which is not identically zero.

a) Show that ω has no zeroes.

b) Show that the complex numbers $\int_a \omega, \int_b \omega \in \mathbb{C}$ are $\mathbb{R}$ -linearly independent.

12. Let X be a compact Riemann surface of genus zero, and suppose there is a meromorphic 1-form ω on X such that ω has only one pole on X , and the pole is a double pole with residue zero. Show that X is biholomorphic to $\hat{\mathbb{C}}$.